

Section 4.3

Let (X, d) be a Metric space, and let $f: X \rightarrow \mathbb{R}$ be a Continuous map.

Prove the zero set of f

$$Z(f) := \{x \in X \mid f(x) = 0\}$$

is closed.

Proof. Let $y \in \overline{Z(f)}$ be given. That is, for any $r > 0$,

$$B_r(y) \cap Z(f) \neq \emptyset.$$

Given $\epsilon > 0$, there exists $\delta > 0$ s.t.

$$d(x, y) < \delta \Rightarrow |f(x) - f(y)| < \epsilon$$

For fixed $\delta > 0$, choose

$$x \in B_\delta(y) \cap Z(f)$$

Then $d(x, y) < \delta$ and $f(x) = 0$. Therefore

$$|f(y) - f(x)| = |f(y)| < \epsilon$$

Since $\epsilon > 0$ was arbitrary, $f(y) = 0$. That is, $y \in Z(f)$.

Alternatively, $Z(f) = f^{-1}[\{0\}]$.

Section 4.2

Let X, Y be Metric spaces. If $f: X \rightarrow Y$ is continuous, then for any $E \subset X$,

$$f[\overline{E}] \subseteq \overline{f[E]}.$$

Proof. Let $f(x) \in f[\overline{E}]$ be given. Since f is continuous, given $\epsilon > 0$, there exists $\delta > 0$ s.t.

$$t \in B_\delta(x) \Rightarrow f(t) \in B_\epsilon(f(x))$$

Meanwhile, since $x \in \overline{E}$, $B_\delta(x) \cap E \neq \emptyset$

Choose $u \in B_\delta(x) \cap E$. Then, $f(u) \in B_\epsilon(f(x))$ and $f(u) \in f[E]$. That is

$$f(u) \in B_\epsilon(f(x)) \cap f[E] \neq \emptyset.$$

This implies $f(x) \in \overline{f[E]}$.

Section 4.4

Let (X, d_X) and (Y, d_Y) be a Metric space, and $f, g: X \rightarrow Y$ be a continuous map.

If $E \subseteq X$ is a dense in X , then $f[E]$ is dense in $f[X]$.

Moreover, If $f|_E = g|_E$, then $f = g$.

Proof. Since f is continuous, $f[X] = f[\overline{E}] \subseteq \overline{f[E]}$ is clear.

Let $x \in X = \overline{E}$ be given. If $f(x) = g(x)$, there is nothing to prove.

Suppose $f(x) \neq g(x)$. Take $\varepsilon = d(f(x), g(x)) > 0$. Since f and g are continuous, there exist $\delta > 0$ s.t.

$$t \in B_\delta(x) \Rightarrow d(f(x), f(t)), d(g(x), g(t)) < \frac{\varepsilon}{2}.$$

Since E is dense in X , there is $t \in B_\delta(x) \cap E$. Now,

$$d(f(x), g(x)) \leq d(f(x), f(t)) + \underbrace{d(f(t), g(t))}_{=0} + d(g(t), g(x)) < \varepsilon = d(f(x), g(x))$$

It is Contradiction.

Thm. Let $f: X \rightarrow Y$ be a map defined on a Compact Metric space X , and any metric space Y .

If f is continuous, then it is continuous uniformly.

Proof. Suppose f is continuous. Given $\epsilon > 0$, for any $x \in X$, there exists $\delta_x > 0$ s.t

$$t \in B_{\delta_x}(x) \Rightarrow f(t) \in B_{\frac{\epsilon}{2}}(f(x))$$

Since the set $\{B_{\frac{\delta_x}{2}}(x) \mid x \in X\}$ is open covering of X and X is Compact, there exists a finite subcover of X :

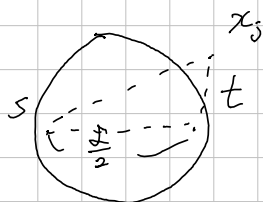
$$\{B_{\frac{\delta_i}{2}}(x_i) \mid i=1, 2, \dots, n\} \quad \text{where } \delta_i = \delta_{x_i}.$$

Take $\delta = \min\{\delta_1, \dots, \delta_n\}$. Choose $s, t \in X$ s.t $d(s, t) < \frac{\delta}{2}$. (Then, $\delta_i + \delta \leq 2\delta_i, \forall i$)
Since the existence of finite subcover, for some $j \leq n$

$$d(t, x_j) < \frac{\delta_j}{2}, \text{ which implies } d(s, x_j) \leq d(s, t) + d(t, x_j) < \frac{\delta}{2} + \frac{\delta_j}{2} \leq \delta_j$$

Finally,

$$d(f(s), f(t)) \leq d(f(s), f(x_j)) + d(f(x_j), f(t)) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$





Section 4, 7. Define

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}: (x, y) \mapsto \begin{cases} 0 & \text{if } (x, y) = (0, 0) \\ \frac{xy^2}{x^2+y^4} & \text{otherwise} \end{cases}, \quad g: \mathbb{R}^2 \rightarrow \mathbb{R}: (x, y) \mapsto \begin{cases} 0 & \text{if } (x, y) = (0, 0) \\ \frac{xy^2}{x^2+y^6} & \text{otherwise} \end{cases}$$

Show that: f is bounded on $\mathbb{R}^2$, g is not bounded in every neighborhood of $(0, 0)$,
 f is not continuous at $(0, 0)$.

The restrictions of f, g on any straight line in $\mathbb{R}^2$ are continuous.

Proof.

g is unbounded, because: for each $x \neq 0$, take $y = x^{\frac{1}{3}}$. Then,

$$g(x, y) = \frac{x \cdot x^{\frac{2}{3}}}{x^2 + x^{\frac{2}{3}}} = \frac{x^{\frac{5}{3}}}{2x^2} = \frac{1}{2} \cdot \frac{1}{x^{\frac{1}{3}}} \rightarrow \infty \text{ as } x \rightarrow 0.$$

Therefore unbounded.

$$2\sqrt{xy} \leq x+y \quad (\sqrt{x}-\sqrt{y})^2 = x+y-2\sqrt{xy}$$

f is bounded, because: $2\sqrt{x^2 y^4} = 2|x| \cdot |y|^2 \leq x^2 + y^4$, that is

$$\left| \frac{xy^2}{x^2+y^4} \right| \leq \frac{1}{2}. \quad (\text{But, how can I find the bound in general setting?})$$

To show the discontinuity of f at $(0, 0)$, take for each $x \neq 0$ $y = \sqrt{x}$. Then,

$$f(x, y) = \frac{xy^2}{x^2+y^4} = \frac{x^2}{x^2+x^2} = \frac{1}{2} \neq f(0, 0).$$

Therefore, $(\frac{1}{n}, \frac{1}{\sqrt{n}}) \rightarrow (0, 0)$, But $f(\frac{1}{n}, \frac{1}{\sqrt{n}}) \rightarrow \frac{1}{2}$, contradiction.

To show the last assertion, if $x = c$ for some $c \in \mathbb{R}$,

$$y \mapsto \frac{cy^2}{c^2+y^4} \text{ is clearly continuous, similarly } y \mapsto \frac{cy^2}{c^2+y^6}.$$

If $y = ax$ for some $a \in \mathbb{R}$, then

$$x \mapsto \frac{a^2 x^3}{x^2 + a^4 x^8} \quad \text{and} \quad x \mapsto \frac{a^2 x^3}{x^2 + a^6 x^{12}} \quad \text{are continuous.}$$

Section 4.8 Let $f: E \rightarrow \mathbb{R}$ where $E \subseteq \mathbb{R}$ be a continuous uniformly map.

If E is bounded, then f is bounded. It is false when E is unbounded.

Proof. Let $M > 0$ be a bound of E . There is, $|a| < M$ for all $a \in E$.

Given $\epsilon > 0$, there exists $\delta > 0$ s.t.

$$|x - y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon$$

By archimedean property, there exists $N \in \mathbb{N}$ s.t. $M < \frac{\delta}{2} \cdot N$

Consider the "Cover" of E :

$$C_k \stackrel{\text{def}}{=} \left(-\frac{\delta}{2}, \frac{\delta}{2}\right) + \frac{\delta}{2}k \quad (k = 0, \pm 1, \dots, \pm(N-1))$$

Then, $\forall x, y \in C_k \cap E$, $|f(x) - f(y)| < \epsilon$ for any k . Finally,

$$\sup\{|f(x)| \mid x \in E\} \leq \max\left\{\sup_{x \in C_k \cap E} |f(x)| \mid k = \pm 0, \dots, \pm(N-1), C_k \cap E \neq \emptyset\right\} < \infty$$

Section 4.9. Let X, Y be Metric spaces.

$f: X \rightarrow Y$ is continuous uniformly if and only if

For any $\epsilon > 0$, there exists $\delta > 0$ s.t. for all $\emptyset \neq E \subseteq X$ satisfying $\text{diam } E < \delta$, $\text{diam } f[E] < \epsilon$

Proof. Suppose f is continuous uniformly. Given $\epsilon > 0$, there exists $\delta > 0$ s.t.

$$\forall x, y \in X, d(x, y) < \delta \Rightarrow d(f(x), f(y)) < \epsilon/2$$

Now, for any non-empty set E satisfying $\text{diam } E < \delta$,

$$d(f(x), f(y)) < \frac{\epsilon}{2} \text{ for all } x, y \in E$$

This implies that $\text{diam } f[E] \leq \frac{\epsilon}{2} < \epsilon$. The converse is trivial, because

$$d(x, y) < \delta \Rightarrow d(f(x), f(y)) = \text{diam } f[\{x, y\}] < \epsilon.$$

Section 4.14 Let $I = [0, 1] \subseteq \mathbb{R}$

If $f: I \rightarrow I$ is continuous, then $f(x) = x$ for some $x \in I$.

Proof. Since f is continuous,

$$h: I \rightarrow \mathbb{R} : x \mapsto f(x) - x$$

is continuous. Clearly, $h(0) = f(0) - 0 \geq 0$ and $h(1) = f(1) - 1 \leq 0$.

Now, by the IVT, we obtain the result.

¶ If $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous uniformly, then there exist $a, b > 0$ s.t.

$$|f(x)| \leq a \cdot |x| + b \quad \text{for all } x \in \mathbb{R}.$$

Proof. There exists $\delta > 0$ s.t. $|x - y| \leq \delta \Rightarrow |f(x) - f(y)| < 1$, by uniform continuity.

Take $a = \frac{1}{\delta}$ and $b = |f(0)| + 1$. Given $x \in \mathbb{R}$, there exists $N \in \mathbb{N}$ s.t. $N\delta < |x| \leq (N+1)\delta$.

$$\begin{aligned} |f(x)| &\leq |f(x) - f(N\delta)| + |f(N\delta) - f((N-1)\delta)| + \dots + |f(\delta) - f(0)| + |f(0)| \\ &< N + |f(0)| + 1 \\ &\leq \frac{1}{\delta}|x| + (|f(0)| + 1) = a|x| + b. \quad \text{gives the result.} \end{aligned}$$

¶ Section 4.0. If $f: X \rightarrow Y$ is not continuous uniformly, then

for some ε_0 , there exist $\{p_n\}, \{q_n\}$ in X satisfying $d(p_n, q_n) \rightarrow 0$ but $d(f(p_n), f(q_n)) > \varepsilon_0$.

Proof. Suppose that given ε_0 , if $\{p_n\}, \{q_n\}$ in X satisfies $d(p_n, q_n) \rightarrow 0$ then $d(f(p_n), f(q_n)) < \varepsilon_0$.

¶ Section 4.11. Let X, Y be Metric spaces.

Suppose that $f: X \rightarrow Y$ is continuous uniformly.

If $\{x_n\}$ is Cauchy sequence, then $\{f(x_n)\}$ is Cauchy sequence.

Proof. Let $\{x_n\}$ be a Cauchy sequence. Given $\varepsilon > 0$, there exists $\delta > 0$ s.t.

$$d(x, y) < \delta \Rightarrow d(f(x), f(y)) < \varepsilon$$

Since $\{x_n\}$ is Cauchy, there exists $N \in \mathbb{N}$ s.t.

$$m, n \geq N \Rightarrow d(x_m, x_n) < \delta$$

Therefore, for $m, n \geq N$, $d(f(x_m), f(x_n)) < \varepsilon$.

Section 4.18. Define the Thomae function as:

$$f: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \begin{cases} 0 & x \notin \mathbb{Q} \text{ or } x=0 \\ \frac{1}{n} & x \in \mathbb{Q} \setminus \{0\}, x = \frac{m}{n} \text{ where } \gcd(m,n)=1. \end{cases}$$

Then, f is continuous at every irrationals, and

f has a simple-discontinuity at every rationals.

Proof. Let $x \in \mathbb{R} \setminus \mathbb{Q}$ be given. Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $1/N < \varepsilon$.

Observe that $S = f^{-1}[\{\frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{N}\}]$ is a finite set, because

for any $n \in \mathbb{N}$, $f(x) = \frac{1}{n}$ if and only if $x = \frac{1}{n}$ or $\frac{2}{n}$ or $\dots$ or $\frac{n-1}{n}$.

Take $\delta = \min\{|x-a| \mid a \in S\}$. Then, this $\delta > 0$ because $x \notin S$ and S is finite.

Now, $|x-t| < \delta \Rightarrow |f(x) - f(t)| = |f(t)| < \frac{1}{N} < \varepsilon$, therefore continuous.

But, unfortunately, at rationals, f is not continuous, because:

Let $x \in \mathbb{Q} \setminus \{0\}$ be given. Denote $x = \frac{m}{n}$ as irreducible form. Take $\varepsilon = \frac{1}{n}$.

For any $\delta > 0$, $B_\delta(x)$ must contain an irrational element. That is,

$$\text{for some } t \in B_\delta(x), |f(x) - f(t)| = \frac{1}{n} \not< \varepsilon$$

Thus, f is not continuous at every rationals, except $x=0$.

But the side-limits exist, because for any sequence $\{x_n\}$ in (x, ∞) s.t. $x_n \rightarrow x$,

$\lim_{n \rightarrow \infty} f(x_n) = 0$ because: for any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $\frac{1}{N} < \varepsilon$ and

except finite terms of $f(x_n)$, $f(x) \leq \frac{1}{N}$. Thus $f(x^+)$ exists, similarly $f(x^-)$ exists.

Section 4.19.

If $f: I \rightarrow \mathbb{R}$ satisfies Intermediate Value Property and for any rational $r \in \mathbb{Q}$, $f^{-1}[\{r\}]$ is closed, then f is continuous. (IVP property: If $f(a) < c < f(b)$, then $\exists x \in (a, b)$ s.t. $c = f(x)$)

Proof. Suppose that f is not continuous w.r.p. Then, there exists a sequence $\{x_n\}$ which converges to x_0 but $f(x_n)$ not converges to $f(x_0)$. That is, there exists $\varepsilon > 0$ s.t.

there exist infinitely many $n \in \mathbb{N}$ satisfying $|f(x_n) - f(x_0)| \geq \varepsilon$. $\left(\Leftrightarrow \begin{array}{l} f(x_0) \leq f(x_n) - \varepsilon < f(x_n) \\ \text{or} \\ f(x_0) \geq f(x_n) + \varepsilon > f(x_n) \end{array} \right)$

By the density of rationals, there exist two rationals $r^- \in (f(x_0) - \varepsilon, f(x_0))$ and $r^+ \in (f(x_0), f(x_0) + \varepsilon)$.

if $f(x_0) < f(x_{n_i})$, then $f(x_0) < r^+ < f(x_{n_i})$ and

if $f(x_{n_i}) < f(x_0)$, then $f(x_{n_i}) < r^- < f(x_0)$.

By the IVP, for each n_i , there exists $t_i \in (x_0, x_{n_i}) \cup (x_{n_i}, x_0)$ s.t. $f(t_i) = r^+$ or r^- .

Since $x_{n_i} \rightarrow x_0$ as $i \rightarrow \infty$, $t_i \rightarrow x_0$ as $i \rightarrow \infty$.

For each $i \in \mathbb{N}$, $t_i \in f^{-1}[\{r^+\}] \cup f^{-1}[\{r^-\}]$, closed set, therefore the limit point x_0 satisfies

$$x_0 \in f^{-1}[\{r^+\}] \cup f^{-1}[\{r^-\}].$$

That is, $f(x_0) = r^+$ or r^- , it is contradiction.

Section 4.20. For non-empty subset $E \subset X$ of a metric space X , define

$$P_E: X \rightarrow \mathbb{R}: x \mapsto \inf_{z \in E} d(x, z)$$

a). $P_E(x) = 0$ if and only if $x \in \bar{E}$.

b). P_E is continuous uniformly.

Proof. $x \in \bar{E} \Leftrightarrow \forall \varepsilon > 0, B_\varepsilon(x) \cap E \neq \emptyset$

$\Leftrightarrow \forall \varepsilon > 0$, there exists $y \in E$ s.t. $d(x, y) < \varepsilon$

By definition, $P_E(x)$ must be smaller than $d(x, y)$ for all $y \in E$, thus smaller than ε .

Therefore $P_E(x) = 0$. To show the uniform continuity, observe that: for any $x, y \in E$,

$$P_E(x) \leq d(x, z) \leq d(x, y) + d(y, z) \text{ for all } z \in E$$

which implies that $P_E(x) \leq d(x, y) + P_E(y)$. Now,

$$|P_E(x) - P_E(y)| \leq d(x, y) \text{ for all } x, y \in E.$$

Given $\varepsilon > 0$, take $\delta = \varepsilon$. Then,

$$d(x, y) < \delta = \varepsilon \Rightarrow |P_E(x) - P_E(y)| \leq d(x, y) < \varepsilon.$$

21. If K, F are ^{disjoint} subset of a metric space X , and K is compact and F is closed, then there exists $\delta > 0$ s.t. for all $x \in K, y \in F$, $d(x, y) \geq \delta$.

Proof. Since K and F are closed and disjoint, $P_F(x) > 0$ for all $x \in K (= \bar{K} \subseteq \bar{F}^c = F^c)$

And, since P_F is continuous, $P_F|_K$ has minimum in K . Therefore, $\inf P_E[K] > 0$.

Counterexample if K is closed but not compact:

In $\mathbb{R}^2$ let $K = \{(x, y) \in \mathbb{R}^2 \mid y \geq e^x\}$ and $F = \{(x, y) \in \mathbb{R}^2 \mid y \leq 0\}$

Let $A \subseteq \mathbb{R}^d$ be a non-empty. define

$$d(\cdot, A): \mathbb{R}^d \rightarrow \mathbb{R}: x \mapsto \inf_{y \in A} \|x - y\|$$

Prove

$$\overline{A} = \bigcap_{n=1}^{\infty} \left\{ x \in \mathbb{R}^d \mid d(x, A) < \frac{1}{n} \right\}$$

It is still true when $\mathbb{R}^d$ be replaced to general Metric Space?

Proof. Let $x \in \overline{A}$ be given. Then, for all $n \in \mathbb{N}$,

$$A \cap B_{\frac{1}{n}}(x) \neq \emptyset.$$

That is, for all $n \in \mathbb{N}$, there exists $a \in A$ s.t. $\|x - a\| < \frac{1}{n}$, this implies

$$d(x, A) \leq d(x, a) < \frac{1}{n}$$

Thus $x \in \overline{A}$. Conversely, let $x \in \bigcap_{n=1}^{\infty} \left\{ x \in \mathbb{R}^d \mid d(x, A) < \frac{1}{n} \right\}$.

Let ε_0 be given. Take $n \in \mathbb{N}$ such that $1 < n\varepsilon_0$

Since for all $n \in \mathbb{N}$, $d(x, A) < \frac{1}{n}$, there exist $a \in A$ s.t.

$$d(x, A) \leq d(x, a) < \frac{1}{n}$$

That is, for fixed ε_0 , we can choose $n \in \mathbb{N}$ s.t. $\frac{1}{n} < \varepsilon_0$ and we can find $a \in A$ s.t.

$$a \in A \cap B_{\frac{1}{n}}(x) \subseteq A \cap B_{\varepsilon_0}(x) \neq \emptyset.$$

22. Let $A, B \subseteq X$ be non-empty disjoint closed sets in metric space X .

Show that a map

$$f: X \rightarrow \mathbb{R}: x \mapsto \frac{P_A(x)}{P_A(x) + P_B(x)}$$

is continuous for $f(x) = 0$ iff $x \in A$ and $f(x) = 1$ iff $x \in B$.

Proof Since there is no element in X s.t.

$$P_A(x) + P_B(x) = 0$$

because it is equivalent to $P_A(x) = P_B(x) = 0$, but it implies that

$$x \in \overline{A} \cap \overline{B} = A \cap B = \emptyset$$

by assumption, which is contradiction. Therefore this f is well-defined.

The continuity is clear. And, observe that

$$\frac{P_A(x)}{P_A(x) + P_B(x)} = 0 \text{ iff } P_A(x) = 0 \text{ iff } x \in A = \overline{A}$$

and

$$\frac{P_B(x)}{P_A(x) + P_B(x)} = 1 \text{ iff } P_B(x) = 0 \text{ iff } x \in B = \overline{B}$$

Every closed subset $A \subseteq X$ is $Z(f)$ for some continuous map $f: X \rightarrow \mathbb{R}$.

Proof. If $A = X$, then put $f = 0$. Suppose that $A \subsetneq X$.

Then, there exists non-empty closed set $B \subset X$ s.t. $A \cap B = \emptyset$. (For example, $B = \{b\}$, $b \notin A$)

Now, $f(x) = \frac{P_A(x)}{P_A(x) + P_B(x)}$, then $Z(f) = A$.

Moreover, observe that

$$A = f^{-1}[\{0\}] \subseteq f^{-1}[[0, \frac{1}{2}]] \text{ and } B = f^{-1}[\{1\}] \subseteq f^{-1}[[\frac{1}{2}, 1]]$$

implies that A and B are covered by disjoint open sets.

Another Question

1. If f, g are continuous uniformly, then fg, fg is?